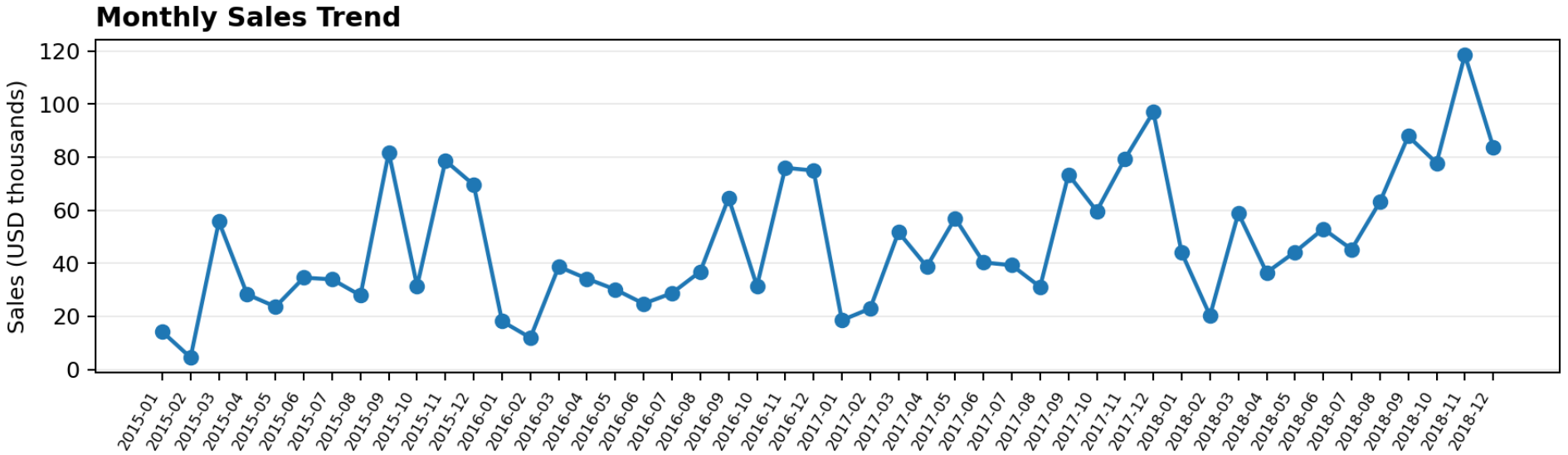
SUPERSTORE RETAIL | EXECUTIVE SALES & PROFIT DASHBOARD

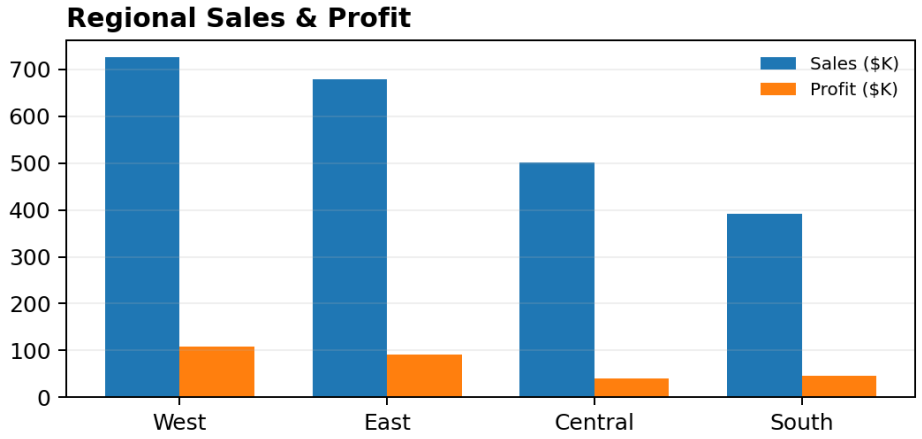
Datheon Internship • Task 2: Interactive Power BI Dashboard – Executive Report

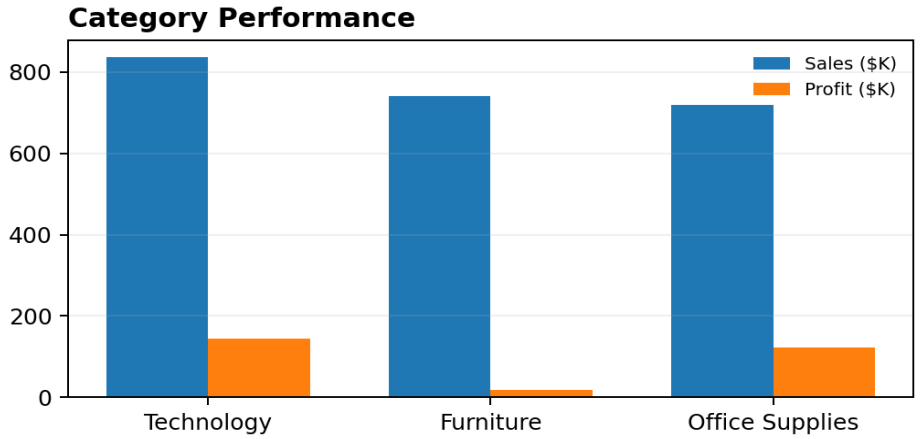
Executive snapshot

$2.30M	$286.4K	5,009	793	12.5%
Total sales	Total profit	Distinct orders	Distinct customers	Profit margin

Sales trend and performance mix



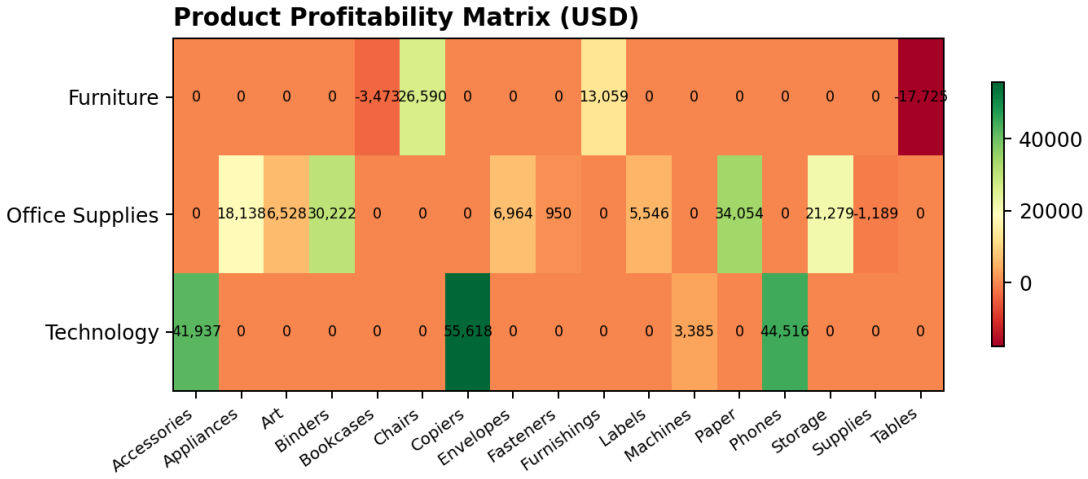


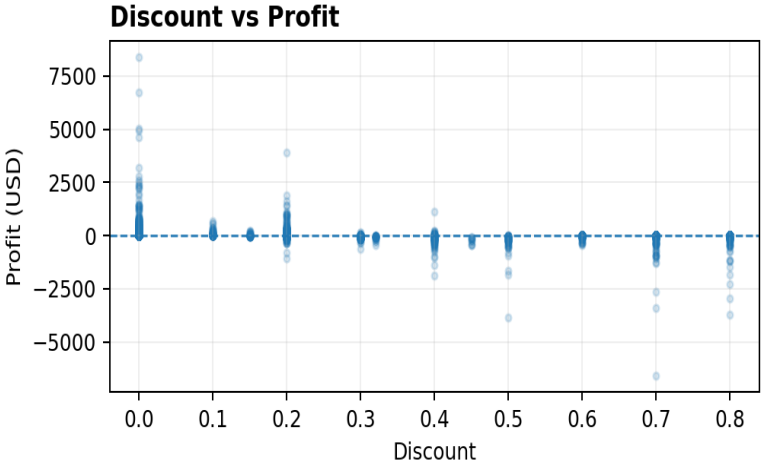


Data scope: 9,994 valid transaction line items, 5,009 distinct orders and 793 customers, covering 2015–2018. Currency is shown as USD, consistent with the source Superstore dataset.

PROFITABILITY & COMMERCIAL DRIVERS

Product profitability and discount sensitivity





Regional performance

Region	Sales (USD)	Profit (USD)	Profit margin
West	$725,458	$108,418	14.9%
East	$678,781	$91,523	13.5%
Central	$501,240	$39,706	7.9%
South	$391,722	$46,749	11.9%

Key observations & recommended actions

1. Revenue scale: Total sales are $2,297,200.86, with $286,397.02 profit across 5,009 distinct orders. Overall profit margin is 12.5%.

2. Category mix: Technology leads category sales ($836,154) and has profit of $145,455. Furniture profit is materially lower relative to its sales than the other categories.

3. Region: West is the largest region by sales ($725,458). Compare regional product mix and discount behavior to identify transferable practices.

4. Customer segment: Consumer is the largest segment by sales ($1,161,401). Use segment-level targeting and retention campaigns while monitoring margin.

5. Discount risk: Discount and profit have a Pearson correlation of -0.219. The relationship is negative in this sample; discount levels of 30% and above frequently show negative average profit. Correlation does not prove causation.

Recommended management focus: Review discount thresholds and loss-making product lines (the lowest-profit sub-category is Tables, with $-17,725 profit). Validate any pricing changes with controlled tests.

POWER BI IMPLEMENTATION NOTES

Recommended report pages and interactions

Page	Recommended visuals	Slicers / interactions
1. Executive Overview	KPI cards: Total Sales, Total Profit, Profit Margin, Orders, Units Sold; monthly sales line chart; sales by region; Category sales/profit.	Order Date range, Region, Segment, Category. Cross-filter all visuals.
2. Product Profitability	Category & Sub-Category treemap; product profitability matrix; top/bottom products by profit; discount vs profit scatter.	Category, Sub-Category, Discount band. Drill down Category → Sub-Category.
3. Customer & Region	Sales/profit by Segment; region comparison; customer/order volume; state-level map if supported.	Region, Segment, State, Order Date.

Suggested star schema

FactOrders: Order ID, Product ID, Customer ID, Order Date, Ship Date, Ship Mode, Sales, Quantity, Discount, Profit. **DimCustomer:** Customer ID, Customer Name, Segment. **DimProduct:** Product ID, Product Name, Category, Sub-Category. **DimDate:** Date, Year, Quarter, Month Number, Month Name. **DimGeography:** Postal Code, City, State, Region, Country. Use one-to-many relationships from dimensions to FactOrders; mark DimDate as the date table.

Core DAX measures

```
Total Sales = SUM(FactOrders[Sales])
Total Profit = SUM(FactOrders[Profit])
Total Orders = DISTINCTCOUNT(FactOrders[Order ID])
Units Sold = SUM(FactOrders[Quantity])
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[Date]))
YoY Sales Growth % = DIVIDE([Total Sales] - [Sales LY], [Sales LY], 0)
```

Data preparation: The supplied CSV contains 10,800 rows. 806 rows have missing/invalid Order Date and are excluded as non-transaction records. The remaining 9,994 rows are retained. Eleven Postal Code values are missing and intentionally left blank rather than imputed.

Deliverable note: This PDF is the exported executive report design and implementation specification. A native .pbix must be created and saved in Microsoft Power BI Desktop; a valid .pbix cannot be authored by this notebook-generation environment.